

UTQIAGVIK NSA, AK, USA

WMO: 700270

Lat: 71.323N

Lon: 156.616W

Elev: 26

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 27516

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-37.5	-33.0	-40.2	0.5	-34.2	-36.7	0.7	-30.9	27.9	-3.0	25.6	-0.4	0.7	N/A	1.039

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	56.1	51.7	52.5	49.7	49.2	47.1	52.2	55.4	49.9	52.5	47.4	49.4	8.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.1	53.7	54.1	47.8	49.1	51.6	45.5	45.0	49.0	21.5	55.7	20.1	52.4	18.8	49.4	58.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.4	22.5	19.9		DB	-41.2	64.9	5.3	4.4	-45.0	68.0	-48.1	70.6	-51.0	73.1	-54.9	76.3
				WB	-38.6	56.6	4.7	2.0	-42.0	58.1	-44.8	59.2	-47.4	60.4	-50.9	61.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	-11.1	-10.7	-11.3	3.4	22.4	34.3	40.8	39.6	34.5	24.2	6.6	-5.2
	DBStd	22.05	12.50	13.34	11.57	10.97	8.11	5.17	5.72	5.41	4.94	6.45	11.00	10.79
	HDD50	13116	1894	1700	1900	1397	856	471	291	328	466	801	1302	1710
	HDD65	18579	2359	2120	2365	1847	1321	921	749	788	916	1266	1752	2175
	CDD50	13	0	0	0	0	0	0	7	6	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	8.3	4.5	5.3	6.6	8.4	9.9	9.9	10.1	9.5	9.9	11.6	9.0	4.7
	PrecAvg	4.4	0.1	0.1	0.1	0.1	0.2	0.3	0.9	1.0	0.7	0.5	0.2	0.2
(15) Precipitation	PrecMax	6.1	0.6	0.4	0.2	0.7	0.4	0.9	2.1	2.6	1.9	1.5	0.5	0.4
	PrecMin	1.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.2	0.1	0.0
	PrecStd	1.2	0.2	0.1	0.1	0.2	0.1	0.2	0.6	0.6	0.4	0.3	0.1	0.1
	PrecStd	1.2	0.2	0.1	0.1	0.2	0.1	0.2	0.6	0.6	0.4	0.3	0.1	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.3	27.5	25.8	33.8	36.3	56.4	63.5	59.9	53.1	37.9	32.7	28.0
		MCWB	28.6	26.7	24.6	32.1	34.6	50.2	55.5	53.5	49.5	36.5	31.5	27.3
	2%	DB	22.1	21.8	18.8	27.6	34.4	50.0	56.8	55.5	47.8	35.0	29.7	21.6
		MCWB	21.3	20.9	18.2	26.7	33.3	46.3	52.5	51.6	45.8	33.4	28.7	20.6
	5%	DB	11.5	15.1	10.9	23.5	33.3	45.0	52.9	52.1	44.4	33.3	26.3	16.5
		MCWB	10.7	14.3	10.3	23.0	32.4	42.3	50.2	49.7	43.3	31.9	25.0	15.9
	10%	DB	5.2	8.6	4.9	19.1	32.4	41.2	49.9	48.4	41.2	32.0	22.6	11.3
		MCWB	4.6	8.0	4.3	18.3	31.6	39.2	47.9	46.6	40.2	30.9	21.5	10.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.4	28.1	27.3	30.7	35.5	51.5	57.0	53.9	49.5	37.7	32.4	27.6
		MCDB	30.2	28.6	28.3	32.4	36.0	56.8	62.4	57.4	52.8	39.2	33.0	28.2
	2%	WB	22.5	23.3	20.2	25.4	34.0	46.8	53.2	51.4	46.2	34.6	30.1	19.6
		MCDB	23.3	24.1	21.0	26.0	34.5	50.4	56.2	54.3	47.7	35.3	30.9	20.4
	5%	WB	13.4	18.9	14.4	21.8	33.0	43.2	50.8	48.5	43.4	32.9	26.6	15.5
		MCDB	14.3	19.7	15.1	22.3	33.6	46.0	53.5	50.4	44.5	33.8	27.7	16.0
	10%	WB	6.6	12.5	6.9	17.2	32.2	40.0	48.8	45.6	40.4	31.7	23.7	9.3
		MCDB	7.1	13.2	7.4	17.8	32.8	41.8	51.1	47.0	41.2	32.7	24.8	9.9
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	12.3	12.2	12.3	7.0	7.4	9.2	7.0	5.4	6.6	10.3	11.6
		MCDBR	20.5	20.0	16.7	12.4	6.7	16.2	16.0	14.8	11.4	6.4	11.9	15.2
		MCWBR	19.9	19.7	16.6	10.4	5.6	12.9	12.1	10.7	8.2	5.9	12.0	15.2
	5% WB	MCDBR	21.4	20.1	15.9	10.7	5.8	17.2	15.4	13.9	9.6	6.9	11.4	15.3
		MCWBR	21.2	19.8	15.8	10.6	5.4	13.4	12.4	11.0	7.9	6.5	11.3	15.0
(40) Clear-Sky Solar Irradiance	taub		N/A	0.201	0.244	0.305	0.315	0.312	0.333	0.329	0.284	0.202	N/A	N/A
	taud		N/A	1.971	2.155	2.049	2.127	2.296	2.418	2.520	2.555	2.135	N/A	N/A
	Ebn at Noon		0	148	239	254	270	277	266	253	234	151	0	0
	Edh at Noon		0	11	22	36	38	33	29	23	16	9	0	0
(44) All-Sky Solar Radiation	RadAvg		0	87	584	1298	1767	2053	1609	886	418	102	0	0
	RadStd		0	10	34	84	146	102	178	89	35	14	0	0

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.99	N/A	N/A	N/A	N/A	N/A	-1147	-1139	N/A	N/A
(47) Regional (3 neighbors)		+2.63	+4.82	+2.48	N/A	N/A	N/A	-869	-873	N/A	N/A

Nomenclature: See separate page